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CSPB 4502 Data Mining
Project Part 2: Proposal Paper

Chess Insights: Result Predictors and Play Sequences by Player Ranking

Problem Statement / Motivation

This project will investigate which features of online chess games are associated with wins, losses, and draws, and whether those relationships change across player rating levels. The project builds on my initial proposal, which identified Lichess.org game data as a strong source for studying game characteristics, opening choices, move sequences, player ratings, and outcomes at large scale. In the initial work, I confirmed that the February 2026 Lichess standard rated games dataset contains the attributes needed to answer these questions, including game metadata, player ratings, results, and move sequences.

The main knowledge I hope to gain is whether common chess advice and assumptions about openings, rating differences, and move patterns are visible in a very large real-world dataset. For example, a move sequence may appear strong overall but may only be successful at certain rating levels, or a predictor such as rating difference may dominate outcome prediction so strongly that opening choice has limited predictive value. This project will also help me apply data mining methods to a dataset that is large, semi-structured, and computationally challenging. The final result will be a reproducible analysis pipeline and a written report that explains which predictors are most associated with game outcomes, which opening or early-move patterns appear most common, and how these findings vary by player ranking.

Literature Survey

Previous chess data studies show that online chess platforms can support large-scale analysis of player behavior, game outcomes, and opening choices. De Marzo and Servedio used Lichess data to study chess openings as a relatedness network, measuring similarity between openings and using online community data to understand the structure and complexity of opening theory. Their work is relevant because it treats openings not only as isolated labels, but as patterns that can be compared and grouped based on how they are played.

Chowdhary et al. used large-scale Lichess data to quantify human performance in chess. Their study notes that the Lichess dataset includes detailed information about played games, including moves, openings, win/loss outcomes, and player skill level. This supports the feasibility of my project because the same general data source contains the features needed to connect player rating, move behavior, and game result.

Recent machine learning work has also explored prediction of chess game outcomes. Samara developed and compared machine learning models for classifying chess game results as White win, Black win, or draw using Lichess data, player rating information, and game metadata. This is closely related to my predictive modeling goal, but my project will place more emphasis on interpretable aggregate comparisons, opening and move-sequence patterns, and differences across rating bins rather than only model performance.

Proposed Work

The project will proceed in several stages: data collection, parsing, cleaning, feature engineering, descriptive analysis, move-sequence analysis, predictive modeling, and evaluation.

First, I will use the February 2026 Lichess standard rated games dataset. The raw file is provided in PGN format compressed as a .pgn.zst file. PGN is a plain-text chess game notation format, so the first major preprocessing step will be parsing the text into a structured format suitable for analysis. I plan to create one main game-level table and, if feasible, a second move-sequence table. The game-level table will contain one row per game, including attributes such as result, White rating, Black rating, rating difference, time control, opening information when available, number of moves, termination type, and selected early moves. The move-sequence table will support analysis of common k-move patterns, such as the first 2, 4, or 6 plies of each game.

Cleaning will include removing or flagging games that are empty, incomplete, improperly parsed, missing required metadata, or have invalid results or ratings. I will standardize outcome labels into White win, Black win, and draw. I will also create a binary perspective-based outcome variable when useful, such as whether the higher-rated player won. This will make it possible to examine both direct result prediction and rating-adjusted result patterns.

Feature engineering will include derived variables such as rating difference, average player rating, rating bin, game length in moves, opening family or first move, first k-move sequence, time-control category, and result from White's perspective. Rating bins will be defined using ranges such as under 1000, 1000-1499, 1500-1999, and 2000+, though I may adjust these ranges after examining the distribution of ratings. Because the full dataset is large, I will first create a smaller random sample for code testing and exploratory analysis. After validating the pipeline, I will run the analysis on the full dataset.

The analysis will include three main components. First, descriptive analysis will summarize the dataset and compare outcome rates by rating bin, rating difference, opening move, time control, and game length. Second, frequent-pattern and association-rule methods will be used to identify common early move sequences and evaluate whether they are associated with particular outcomes. Third, supervised classification models will test whether game metadata and engineered features can predict outcome categories. I will begin with interpretable baseline models, such as logistic regression or decision trees, and may compare them with random forest models if time allows.

My work differs from previous studies in its scope and emphasis. Rather than focusing only on opening similarity, only human performance, or only predictive accuracy, this project will combine interpretable descriptive analysis, frequent sequence analysis, and classification. It will also explicitly compare results across player rating bins, which should help determine whether predictors of success differ between lower-rated, intermediate, and higher-rated players.

Dataset

The dataset is the Lichess dataset from February 2026. I downloaded it and confirmed that my proposed questions can be answered with the data. The file was 27.8GB compressed, and 184.12GB decompressed. It contains information for 84,600,043 chess games. The dataset citation is provided in the works cited section. This dataset is relevant because it contains the key fields needed for the project: game results, player ratings, game metadata, and full move sequences. Its large size also makes it appropriate for a data mining project because it supports analysis across rating groups and less common opening patterns while requiring careful preprocessing and computational planning.

Evaluation Methods

Because the project includes multiple types of analysis, I will use several evaluation methods.

For data description, I will report descriptive statistics such as counts, percentages, means, medians, standard deviations, quartiles, and interquartile ranges. These will be used for player ratings, rating differences, game length, outcome frequencies, and opening frequencies. I will also use grouped summaries to compare results across rating bins.

For association rules and frequent patterns, I will use support, confidence, and lift. Support will measure how common a move sequence or opening pattern is. Confidence will measure how often a result occurs given a particular pattern. Lift will help determine whether a pattern is more strongly associated with an outcome than would be expected from the baseline outcome rate.

For categorical comparisons, I will use chi-squared tests on contingency tables, such as opening category by result or rating bin by result. This will help evaluate whether observed differences in outcome distributions appear meaningful rather than simply descriptive.

For classification, I will use train/test splits and evaluate models with accuracy, precision, recall, F1 score, specificity where appropriate, and confusion matrices. Since chess outcome prediction may involve imbalanced classes, especially if draws are less common than wins/losses in some subsets, I will not rely on accuracy alone. I will compare model performance against simple baselines, such as always predicting the most common class or predicting the higher-rated player to win.

If I use probabilistic models or compare ranking-style predictive performance, I may also report ROC/AUC for binary classification tasks, such as predicting whether White wins or whether the higher-rated player wins. I would use ROC/AUC because it evaluates how well a model separates positive and negative cases across thresholds, which is useful when class probabilities are more informative than hard class labels.

Tools

I plan to use Python as the primary programming language because it supports the full workflow from parsing to modeling and visualization. I will use zstandard or command-line decompression tools to work with .zst files. I will use python-chess to parse PGN files and extract moves, metadata, and game-level information. I will use pandas and NumPy for data cleaning, transformation, sampling, and aggregation. I will store structured output as Parquet files because Parquet is more efficient than CSV for large tabular datasets.

For modeling and evaluation, I will use scikit-learn for train/test splits, classification models, confusion matrices, and performance metrics. For visualization, I will use matplotlib and seaborn to create charts and tables that support the final report. If the full dataset is too large for memory, I may use chunked processing for more efficient querying and aggregation.

Milestones

July 5-July 12: Clean and standardize the parsed data. Create the full analysis pipeline and test it on the smaller sample data. Create derived fields such as rating difference, average rating, rating bin, time-control category, and outcome labels. Run descriptive analysis on the sample dataset. Produce initial tables and visualizations for outcome rates, rating bins, common openings, and game length.

July 12-July 19: Use pipeline development and results of running on the sample dataset to prepare the Progress Report, which is due July 17th.

July 19-July 26: Create the move-sequence table and calculate support, confidence, and lift for selected early move patterns. Build baseline classification models and evaluate them using accuracy, precision, recall, F1 score, specificity, and confusion matrices. Compare results across rating bins.

July 26-August 2: Expand the analysis to the full dataset. Make any necessary adjustments to the code to process the data in chunks if needed due to size.

August 2-August 9: Evaluate the results of the analysis and modeling and begin to identify the strongest findings.

August 9-August 14: Refine visualizations, interpret results, and document limitations. Finalize Final Report, which is due August 14.

Works Cited

Chowdhary, Sandeep, Iacopo Iacopini, and Federico Battiston. "Quantifying Human Performance in Chess." *Scientific Reports*, vol. 13, 2023, article no. 2113, https://doi.org/10.1038/s41598-023-29369-7.

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Lichess. "Lichess Database: Standard Rated Games, February 2026." *Lichess*, 2026, https://database.lichess.org/.

Samara, Khalid. "Machine Learning Approaches for Classifying Chess Game Outcomes: A Comparative Analysis of Player Ratings and Game Dynamics." *Electronics*, vol. 15, no. 1, 2026, article no. 1, https://doi.org/10.3390/electronics15010001.